STABILITY / UPTIME

⚠ CRITICAL

CASCADING SERVICE FAILURE

One service went unhealthy and took three dependent services down with it. The cascade is spreading. The on-call phone hasn't stopped ringing. Users are posting screenshots of error pages and asking for updates on social media.

Cascading failures are the hallmark of tightly coupled systems with insufficient resilience engineering. The SRE toolkit for preventing them includes: circuit breakers (automatically stopping calls to an unhealthy downstream service), bulkheads (isolating failure so one component cannot exhaust shared resources), retry logic with exponential backoff (preventing retry storms), health check endpoints and readiness probes, and graceful degradation (returning partial data rather than total failure). The Amazon AWS architecture uses cell-based design to limit blast radius — a failure in one cell cannot propagate to others. Runbooks and on-call playbooks must be maintained and rehearsed so that incident response is procedural, not improvised.

Tight coupling is a hidden liability. One failure becomes many when there is no isolation between services.

TEAM DISCUSSION

Do you have circuit breakers, health checks, and runbooks ready — or are you improvising under pressure right now?